BÖHLER Q G 3

Solid Wire, mild steel

Classifications

EN ISO 14341-A	AWS A5.18 / SFA-5.18
G 42 3 M21 3Si1	ER70S-6
G 38 2 C1 3Si1	

Characteristics and typical fields of application

GMAW solid wire for non- and low-alloyed steel.
For applications in structural steel engineering, pressure vessel and ship building.

Base materials

Steels with yield strength < 420 MPa
S235JR-S355JR, S235J0-S355J0, S235J2-S355J2, S275N-S420N, S275M-S420M, P235GH-P355GH, P275NL1-P355NL1, P215NL, P265NL, P355N, P285NH-P420NH, P195TR1-P265TR1, P195TR2-P265TR2, P195GH-P265GH, L245NB-L415NB, L245MB-L415MB, GE200-GE240, Schiffbaustähle: A, B, D, E, A 32-E 36
ASTM A 106 Gr. A, B, C; A 181 Gr. 60, 70; A 283 Gr. A, C; A 285 Gr. A, B, C; A 350 Gr. LF1; A 414 Gr. A, B, C, D, E, F, G; A 501 Gr. B; A 513 Gr. 1018; A 516 Gr. 55, 60, 65, 70; A 573 Gr. 58, 65, 70; A 588 Gr. A, B; A 633 Gr. C; A 662 Gr. B; A 711 Gr. 1013; A 841 Gr. A; API 5 L Gr. B, X42, X52, X56, X60

Typical analysis

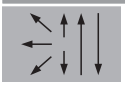
	C	Si	Mn
wt.-%	0.08	0.9	1.45

Mechanical properties of all-weld metal - typical values (min. values)

Condition	Yield strength R_e	Tensile strength R_m	Elongation A (L_0=5d_0)	Impact energy ISO-V KV J		
	MPa	MPa	%	20°C	-20°C	-30°C
u	440 (≥ 420)	530 (500 - 640)	30 (≥ 20)	160 (≥ 47)		≥ 47
u2	420 (≥ 380)	510 (470 - 600)	26 (≥ 20)	120 (≥ 47)	≥ 47	

u untreated, as welded – shielding gas M21
u2 untreated, as welded – shielding gas I1

Operating data



Polarity	DC+
Shielding gas (EN ISO 14175)	C1 M21

Dimension mm
0.8
0.9
1.0
1.2
1.6

Approvals

TÜV (19788), DB (42.132.89), CE

All information provided is based upon careful investigation and intensive research.
However, we do not assume any liability for correctness and information is subject to change without notice.